

COSC 594/650: Optimization for Machine Learning

Homework 3

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This homework is due on **10/2, 11:59pm**. Format your solution in PDF and submit it on Canvas.

1. (**Projection onto a set of linear constraints**) Consider the optimization problem

$$\begin{aligned} & \underset{x \in \mathbb{R}^n}{\text{minimize}} && \frac{1}{2} \|a - x\|_2^2 \\ & \text{subject to} && Ax = b, \end{aligned} \tag{1}$$

which projects the point a onto the set

$$\mathcal{X} \triangleq \{x \in \mathbb{R}^n \mid Ax = b\}.$$

Suppose that A is a fat matrix and AA^T is invertible.

- (a) Write down the Lagrangian function.

Solution:

- (b) Find the dual problem of (1).

Solution:

- (c) Write down the KKT conditions for (1).

Solution:

- (d) Let $[a]_+$ be the optimizer of (1). Using the KKT conditions and the dual optimization problem, express $[a]_+$ as a function of A , b , and a .

Solution:

2. (**SVM and kernels**) Consider the soft-margin SVM optimization problem

$$\underset{w, v}{\text{minimize}} \quad \sum_{i=1}^n \max \{0, 1 - y_i (w^T x_i + v)\} + \lambda \|w\|_2^2, \tag{2}$$

where $\lambda > 0$. This optimization problem is nonsmooth; that is, its objective function is not differentiable.

- (a) Show that the optimizer of (2) can also be obtained by solving the constrained smooth optimization problem

$$\begin{aligned} & \underset{w, \xi, v}{\text{minimize}} && \sum_{i=1}^n \xi_i + \lambda \|w\|_2^2 \\ & \text{subject to} && 1 - y_i (w^T x_i + v) \leq \xi_i, \quad \forall i, \\ & && \xi_i \geq 0, \quad \forall i. \end{aligned} \tag{3}$$

Solution:

- (b) Find the dual problem of (3).

Solution:

- (c) Write the KKT conditions for (3).

Solution:

- (d) Assume that $\{x_i\}_{i=1}^n$ is not known. Instead, suppose that we have access to the kernel matrix K , whose (i, j) -th entry is

$$K_{i,j} = \langle x_i, x_j \rangle.$$

Explain how the optimizer of (2) can be computed in this scenario.

Solution:

3. (**Gradient projection**) Consider the constrained optimization problem

$$\begin{aligned} & \underset{x \in \mathbb{R}^n}{\text{minimize}} && \frac{1}{2} x^T Q x + b^T x \\ & \text{subject to} && 0 \leq x_i \leq 1, \quad \forall i, \end{aligned} \tag{4}$$

where $x = (x_1, \dots, x_n)$.

- (a) Generate Q and b according to the following procedure:

- i. Set $n = 100$.
- ii. Generate a random matrix $B \in \mathbb{R}^{n \times n}$ whose entries are drawn from a standard Gaussian distribution $\mathcal{N}(0, 1)$; that is,

$$B = \text{randn}(100, 100).$$

- iii. Generate a diagonal matrix $D \in \mathbb{R}^{n \times n}$ whose diagonal entries are drawn from a standard log-normal distribution; that is,

$$D = \text{diag}(\exp(\text{randn}(n, 1))).$$

iv. Set

$$Q = BB^T + D.$$

v. Generate a random vector b from a zero-mean Gaussian distribution with variance 100; that is,

$$b = 10 \text{ randn}(n, 1).$$

Solution:

(b) Implement the gradient projection method using both the constant step size

$$\alpha^r = \frac{1}{L}$$

and the diminishing step size

$$\alpha^r = \frac{5 \ln(r)}{rL}.$$

Initialize both algorithms from $x^0 = 0$.

Solution:

(c) Let x^* be the optimal solution of (4). We learned in class that

$$x^* = [x^* - \nabla f(x^*)]_+,$$

where f is the objective function and $[\cdot]_+$ is the projection operator onto the feasible set. Define

$$\theta(x) \triangleq \|x - [x - \nabla f(x)]_+\|_2^2.$$

Plot $\log(\theta(x^r))$ versus the iteration number r for both step-size selection rules over 2000 iterations.

Solution:

4. **(Linear scaling and Newton's method)** Consider the unconstrained optimization problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x). \tag{5}$$

Applying a linear scaling of the variables leads to

$$\underset{z \in \mathbb{R}^n}{\text{minimize}} \quad f(Az), \tag{6}$$

where $A \in \mathbb{R}^{n \times n}$ is invertible. Start from $x^0 = 0$ in (5) and $z^0 = 0$ in (6), and apply Newton's method to both optimization problems in parallel.

- (a) Show that Newton's method is invariant under linear scaling of the variables; that is, show that

$$x^r = Az^r, \quad \forall r.$$

Is this property true for gradient descent?

Solution:

- (b) Let f be a quadratic function. Show that Newton's method converges in one iteration.

Solution: